Method of expressing concentration of solution

1. (c) $M_1V_1 + M_2V_2 = MV$

2. (d) $M = \frac{w}{m \times V(l)}$; $0.25 = \frac{w}{106 \times 0.25}$;$w = 6.625\ gm$

3. (d) $N_1V_1 = N_2V_2$

$2 \times 1 = N_2 \times 6$

$N_2 = 0.33$

4. (e) 5.85 *g NaCl* = $\frac{5.85}{58.5} mole = 0.1mol$

90 *g* $H_2O = \frac{90}{18} moles = 5moles$

mole fraction of *NaCl* = $\frac{0.1}{5+0.1} \approx 0.0196.$

5. (b) $M = \frac{n}{V(l)} = \frac{0.006}{0.1} = 0.06$

6. (b) $M = \frac{W \times 1000}{mol.mass \times \text{Volume in } ml.} = \frac{9.8 \times 1000}{98 \times 2000} = 0.05M$

7. (a) $M = \frac{W}{m.wt.} \times \frac{1000}{\text{Volume in } ml.} = \frac{5 \times 1000}{40 \times 250} = 0.5M$

8. (d) Basicity of H_3PO_3 is 2.

Hence 0.3 *M* $H_3PO_3 = 0.6N$.

9. (d) 2 *gm.* Hydrogen has maximum number of molecules than others.

11. (a) $M_1V_1 = M_2V_2$

$0.01 \times 19.85 = M_2 \times 20$

$M_2 = 0.009925;\ \ M = 0.0099.$

12. (b) 1500 cm^3 of 0.1 *N HCl* have number of *gm* equivalence

$= \frac{N_1 \times V_1}{1000} = \frac{1500 \times 0.1}{1000} = 0.15$

$\therefore 0.15gm.$ equivalent of *NaOH* $= 0.15 \times 40 = 6gm.$

13. (a) $M = \frac{w}{m.wt. \times \text{volume inlitre}} = \frac{5.85}{58.5 \times 0.5} = 0.2M$

14. (a) Molecular weight of $C_2H_5OH = 24 + 5 + 16 + 1 = 46$

Molecular mass of $H_2O = 18$

414*g* of C_2H_5OH has $\frac{414}{46} = 9$ *mole*

18*g* of H_2O has $= \frac{18}{18} = 1$ *mole*

Mole fraction of water $= \frac{n_1}{n_1+n_2} = \frac{1}{1+9} = \frac{1}{10} = 0.1$

15. (b)$17gmNH_3 = 1$ *mole*.

Molecules of $NH_3 = \frac{6.02\times10^{23}\times4.25}{17} = 1.5 \times 10^{23}$

17. (b)$(2.5 \times 1 + 3 \times 0.5) = M_3 \times 5.5$

or $2.5 + 1.5 = M_3 \times 5.5$ or $M_3 = \frac{4}{5.5} = 0.73M$.

20. (b) Normality of $2.3MH_2SO_4 = M \times$Valency

$= 2.3 \times 2 = 4.6N$

1.